

Laboratorio 7

1) $S \rightarrow OAO \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow SIA$

$C \rightarrow SIC$

Eliminar epsilon

$S \rightarrow OAO \mid 1B1 \mid BB \mid OAO$

$S \rightarrow OAO \mid 11 \mid 1B1 \mid B$

$S \rightarrow OAO \mid OAO \mid 1B1 \mid 11 \mid BB \mid B$

$S \rightarrow OAO \mid OAO \mid 1B1 \mid 11 \mid BB \mid B$

$A \rightarrow C$

$B \rightarrow SIA$

$C \rightarrow S$

Eliminar Producciones ~~unarias~~ unarias

$P_0 = \{ (S, S) (A, A) (B, B) (C, C) \}$

$(S, S) \quad S \rightarrow B \quad (S, B)$

$(S, B) \quad B \rightarrow A \quad (S, A)$

$(S, A) \quad A \rightarrow C \quad (S, C)$

$(S, C) \quad C \rightarrow S \quad (S, S)$

$(B, B) \quad B \rightarrow A \quad (B, A)$

$(B, A) \quad A \rightarrow C \quad (B, C)$

$(B, C) \quad C \rightarrow S \quad (B, S)$

$(A, A) \quad A \rightarrow C \quad (A, C)$

$(A, C) \quad C \rightarrow S \quad (A, S)$

$(C, C) \quad C \rightarrow S \quad (C, S)$

$(S, S) \rightarrow 0AO|1B1|00|11|BB$

$(S, A) (S, B) (S, C)$ no generan nuevas opciones

$(B, S) \rightarrow 0AO|1B1|00|11|BB$

$(B, B) (B, A) (B, C)$ no generan nuevas opciones

$(A, S) \rightarrow 0AO|1B1|00|11|BB$

$(A, A) (A, B) (A, C)$ no generan nuevas opciones

$(C, S) \rightarrow 0AO|1B1|00|11|BB$

$(C, C) (C, B) (C, A)$ no generan nuevas opciones

$S \rightarrow 0AO|1B1|00|11|BB$

$A \rightarrow 0AO|1B1|00|11|BB$

$B \rightarrow 0AO|1B1|00|11|BB$

$C \rightarrow 0AO|1B1|00|11|BB$

Eliminar inútiles e inalcanzables

$T = \{0, 1\} \quad \gamma_1 = \{S\}$

$w_1 = \{S, A, B, C\} \quad \gamma_2 = \{S, A, B, 0, 1\}$

$w_2 = \{S, A, B, C\} \quad \gamma_2 = \{S, A, B, 0, 1\}$

$S \rightarrow 0AO|1B1|00|11|BB$

$A \rightarrow 0AO|1B1|00|11|BB$

$B \rightarrow 0AO|1B1|00|11|BB$

Forma de Chomsky

$T_0 = 0 \quad T_1 = 1$

$S \rightarrow T_0 A T_0 | T_1 B T_1 | 00 | 11 | BB$

$A \rightarrow T_0 A T_0 | T_1 B T_1 | 00 | 11 | BB$

$B \rightarrow T_0 A T_0 | T_1 B T_1 | 00 | 11 | BB$

$C_1 = A T_0 \quad C_2 = B T_1$

$$S \rightarrow T_0 C_1 | T_1 C_2 | \infty | 11 | BB$$
$$A \rightarrow T_0 C_1 | T_1 C_2 | \infty | 11 | BB$$
$$B \rightarrow T_0 C_1 | T_1 C_2 | \infty | 11 | BB$$

2) $S \rightarrow aAa | bBb | C$

$$A \rightarrow C | a$$
$$B \rightarrow C | b$$
$$C \rightarrow CDE | C$$
$$D \rightarrow A | B | a | b$$

Eliminar epsilon

$$S \rightarrow aAa | aa | bBb$$
$$S \rightarrow aAa | bb | bBb$$
$$S \rightarrow aAa | aa | bb | bBb$$
$$A \rightarrow C | a$$
$$B \rightarrow C | b$$
$$C \rightarrow CDE | DE | CE | E$$
$$D \rightarrow A | B | a | b$$

Eliminar producciones unarias

$$P_0 = \{(S, S) (A, A) (B, B) (C, C) (D, D)\}$$
$$(C, C) \quad C \rightarrow E \quad (C, E)$$
$$(S, S)$$
$$(A, A) \quad A \rightarrow C \quad (A, C)$$
$$(A, C) \quad C \rightarrow E \quad (A, E)$$
$$(B, B) \quad B \rightarrow C \quad (B, C)$$
$$(B, C) \quad C \rightarrow E \quad (B, E)$$

(D, D) $D \rightarrow A (D, A)$

(D, A) $A \rightarrow C (D, C)$

(D, C) $C \rightarrow E (D, E)$

(D, D) $D \rightarrow B (D, B)$

(D, B)

(C, C) CDE | DE | CE

(C, E) No produce nada

(S, S) aAa | aa | bb | bBb

(A, A) a

(A, C) CDE | DE | CE

(B, B) b

(B, C) CDE | DE | CE

(D, D) ab

(D, A) a

(D, B) b

(D, C) CDE | DE | CE

(D, E) No produce nada

$S \rightarrow aAa | aa | bb | bBb$

$A \rightarrow CDE | DE | CE | a$

$B \rightarrow CDE | DE | CE | b$

$C \rightarrow CDE | DE | CE$

$D \rightarrow CDE | DE | CE | ab | a | b$

Eliminar inútiles o inalcanzables

$$T = \{a, b\}$$

$$w_1 = \{S, A, B, D, a, b\} \quad \gamma_1 = \{S\}$$

$$w_2 = \{S, A, B, C, D, a, b\} \quad \gamma_2 = \{D, A, B, a, b, S\}$$

$$w_3 = \{S, A, B, C, D, a, b\} \quad \gamma_3 = \{D, A, B, a, b, C, S\}$$

$$S \rightarrow aAa \mid aa \mid bb \mid bBb$$

$$A \rightarrow CDE \mid DE \mid CE \mid a$$

$$B \rightarrow CDE \mid DE \mid CE \mid b$$

$$C \rightarrow CDE \mid DE \mid CE$$

$$D \rightarrow CDE \mid DE \mid CE \mid ab \mid a \mid b$$

Forma de Chomsky

$$C_1 = DE \quad T_1 = a \quad T_2 = b$$

$$S \rightarrow T_1 A T_1 \mid T_1 T_1 \mid T_2 T_2 \mid T_2 B T_2$$

$$A \rightarrow CC_1 \mid DE \mid CE \mid T_1$$

$$B \rightarrow CC_1 \mid DE \mid CE \mid T_2$$

$$C \rightarrow CC_1 \mid DE \mid CE$$

$$D \rightarrow CC_1 \mid DE \mid CE \mid T_1 T_2 \mid T_1 \mid T_2$$

$$C_2 = AT_1 \quad C_3 = BT_2$$

$$S \rightarrow T_1 C_2 \mid T_1 T_1 \mid T_2 T_2 \mid T_2 C_3$$

$$A \rightarrow CC_1 \mid DE \mid CE \mid T_1$$

$$B \rightarrow CC_1 \mid DE \mid CE \mid T_2$$

$$C \rightarrow CC_1 \mid DE \mid CE$$

$$D \rightarrow CC_1 \mid DE \mid CE \mid T_1 T_2 \mid T_1 \mid T_2$$

$$3) S \rightarrow ASA | aB$$

$$A \rightarrow B | S$$

$$B \rightarrow b | \epsilon$$

Eliminar epsilon

$$S \rightarrow ASA | a | aB$$

$$S \rightarrow SA | AS | ASA | aB$$

$$S \rightarrow ASA | SA | AS | a | aB$$

$$S \rightarrow ASA | SA | AS | aB | a | S$$

$$A \rightarrow B | S$$

$$B \rightarrow b$$

Eliminar producciones unarias

$$S \rightarrow A \rightarrow B$$

$$P_0 = \{ (S, S) (A, A) (B, B) \}$$

$$(A, A) \quad A \rightarrow B \quad (A, B)$$

$$(A, A) \quad A \rightarrow S \quad (A, S)$$

$$(A, A) \quad \text{No aporta}$$

$$(A, S) \quad ASA | AS | SA | aB | a$$

$$(A, B) \quad b$$

$$S \rightarrow ASA | AS | SA | aB | a$$

$$A \rightarrow ASA | AS | SA | aB | a | b$$

$$B \rightarrow b$$

Eliminar inútiles e inalcanzables

$$T_1 = \{a, b\}$$

$$\gamma_1 = \{S\}$$

$$w_1 = \{S, A, B, a, b\} \quad \gamma_2 = \{S, A, a, b\}$$

$$w_2 = \{S, A, B, a, b\} \quad \gamma_3 = \{S, A, B, a, b\}$$

$$\gamma_4 = \{S, A, B, a, b\}$$

$$S \rightarrow ASA | AS | SA | aB | a$$

$$A \rightarrow ASA | AS | SA | aB | a | b$$

$$B \rightarrow b$$

Forma de Chomsky

$$C_1 = AS \quad T_1 = a \quad T_2 = b$$

$$S \rightarrow C_1 A | AS | SA | T_1 B | T_1$$

$$A \rightarrow C_1 A | AS | SA | T_1 B | T_1 | T_2$$

$$B \rightarrow T_2$$